

# Guillaume Payeur

## PhD Student in Physics

Université de Montréal — Ciela — Mila — Trottier Space Institute

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### ▷ Research

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I'm a PhD student in physics at the Université de Montréal, under the supervision of Laurence Perreault-Levasseur. I'm also a member of [Mila](#), [Ciela](#) and the [Trottier Space Institute](#).

### ▷ Education

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|-------------------------------------------------------------------------------|------------------------------|
| <b>PhD in Physics</b><br><i>Université de Montréal</i>                        | 2025 – Present               |
| <b>MSc in Physics (High Energy Theory)</b><br><i>McGill University</i>        | 2023 – 2025<br>GPA: 4.0/4.0  |
| <b>BSc in Mathematics &amp; Physics (Honours)</b><br><i>McGill University</i> | 2019 – 2023<br>GPA: 3.98/4.0 |
| <b>DCS in Pure and Applied Science</b><br><i>Champlain College</i>            | 2017 – 2019                  |

### ▷ Publications

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- Guillaume Payeur, Laurence Perreault-Levasseur, Gabriel Missael Barco, and Yashar Hezaveh. Strong gravitational lensing posterior sampling in pixel-space using diffusion models and recurrent inference machines, 2026  
<https://arxiv.org/abs/2607.19459>
- Guillaume Payeur, Evan McDonough, and Robert Brandenberger. Do observations prefer thawing quintessence? *Physical Review D*, 111(12):123541, June 2025  
<https://journals.aps.org/prd/abstract/10.1103/bggr-61nr>
- Guillaume Payeur, Evan McDonough, and Robert Brandenberger. Swampland conjectures constraints on dark energy from a highly curved field space. *Physical Review D*, 110(10):106011, November 2024  
<https://journals.aps.org/prd/abstract/10.1103/PhysRevD.110.106011>
- Edward W. Kolb, Andrew J. Long, Evan McDonough, and Guillaume Payeur. Completely dark matter from rapid-turn multifield inflation. *Journal of High Energy Physics*, 2023(2):181, February 2023  
[https://link.springer.com/article/10.1007/JHEP02\(2023\)181](https://link.springer.com/article/10.1007/JHEP02(2023)181)

1. Guillaume Payeur, Étienne Artigau, Laurence Perreault Levasseur, and René Doyon. Correlated read noise reduction in infrared arrays using deep learning. *The Astronomical Journal*, 163(6):292, may 2022  
<https://iopscience.iop.org/article/10.3847/1538-3881/ac69d2>

## ▷ Talks

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| <b>Do Observations Prefer Thawing Quintessence?</b><br><i>Dark Side of the Universe</i>                | Montreal, 2025 |
| <b>Do Observations Prefer Thawing Quintessence?</b><br><i>McGill HEP Theory Journal Club</i>           | Montreal, 2025 |
| <b>Completely Dark Matter &amp; the de Sitter Swampland Conjecture</b><br><i>WITP Summer Symposium</i> | Winnipeg, 2022 |
| <b>Readout Noise &amp; Machine Learning</b><br><i>IREX End-of-Summer Presentations</i>                 | Montreal, 2020 |

## ▷ Awards & Funding

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|------------------------------------------------------------------|-------------------------|
| <b>NSERC Canada Graduate Scholarship – Doctoral</b><br>\$120,000 | Fall 2025 – Summer 2027 |
| <b>Dr &amp; Mrs Milton Leong Fellowship</b><br>\$5,000           | Fall 2024 – Winter 2025 |
| <b>FRQNT Master’s Training Scholarship</b><br>\$40,000           | Fall 2023 – Winter 2025 |
| <b>NSERC Canada Graduate Scholarship – Master’s</b><br>\$17,500  | Fall 2023 – Winter 2024 |
| <b>Lorne Trottier Fellowship</b><br>\$5,000                      | Fall 2023 – Winter 2024 |
| <b>Trottier Space Institute Graduate Fellowship</b><br>\$2,000   | Fall 2023 – Winter 2025 |
| <b>First Class Honours in Mathematics and Physics</b>            | Winter 2023             |
| <b>Wing Hing Chang Scholarship in Science</b><br>\$800           | Fall 2021 – Winter 2022 |
| <b>NSERC Undergraduate Student Research Award</b><br>\$6,000     | Summer 2021             |
| <b>McGill Dean’s Honour List</b>                                 | Fall 2020 – Winter 2021 |
| <b>Bourse Hors UdeM</b><br>\$4,000                               | Summer 2020             |
| <b>Faculty of Science Scholarship</b><br>\$700                   | Fall 2020 – Winter 2021 |
| <b>McGill Dean’s Honour List</b>                                 | Fall 2019 – Winter 2020 |
| <b>Champlain College Dean’s List</b>                             | Winter 2018             |
| <b>Champlain College Dean’s List</b>                             | Fall 2018               |

## ▷ Teaching

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**Course Teaching Assistant at McGill**  
• PHYS 362 Statistical Mechanics (Winter 2025) — 1 lecture

- PHYS 346 Major Quantum Mechanics (Fall 2024) — 1 lecture, 6 tutorials
- PHYS 142 Electromagnetism and Optics (Winter 2024) — 4 tutorials
- PHYS 251 Honours Classical Mechanics I (Fall 2023) — 1 lecture, 4 tutorials

#### ▷ Other Honours

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**Second Place, Mila Speed Science Contest 2026** 2026  
Mila

**Edith Engelberg Teaching Award Finalist** 2024 – 2025  
McGill University

**Second Place, Astromatic Hackathon** 2022  
Université de Montréal